ABSTRACT

This project presents a Retrieval-Augmented Agentic Framework for generating Business Requirement Documents (BRDs) from heterogeneous enterprise documents. In current enterprise environments, BRD creation is largely a manual and time-intensive process that involves analysing large volumes of unstructured data such as PDFs, Word documents, Excel sheets, and informal client inputs. This often results in inconsistencies, delays, and lack of traceability.

To address these challenges, the proposed system integrates Retrieval-Augmented Generation (RAG) with an agentic workflow to enable intelligent and context-aware document synthesis. The system incorporates document ingestion and preprocessing, semantic embedding generation, vector-based knowledge storage, and retrieval mechanisms to extract relevant information from diverse data sources.

An adaptive query routing mechanism is introduced to handle different types of queries by dynamically selecting the appropriate knowledge source. Depending on the nature of the query, the system retrieves information from the document knowledge base, structured databases, or external web sources. This enhances flexibility and improves the overall quality of responses.

The framework further employs an agent-based reasoning approach, where tasks are divided among components responsible for planning, retrieval, content generation, and validation. This modular design supports structured content generation and maintains logical consistency across outputs.

The primary objectives of this work include reducing the time required for BRD creation, improving accuracy and consistency, and ensuring traceability of generated content to source documents. The system also demonstrates scalability across enterprise domains and provides a foundation for automated documentation workflows.

Overall, the proposed framework contributes toward improving efficiency and reliability in enterprise documentation processes by combining retrieval, reasoning, and adaptive decision-making within a unified system.

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1. INTRODUCTION AND BACKGROUND

1.1 Introduction

The rapid growth of enterprise data has led to an increasing need for intelligent systems capable of extracting meaningful insights from unstructured documents. Organizations today manage large volumes of heterogenous data sources such as PDFs, Word documents, spreadsheets, and internal knowledge repositories. Manual extraction of relevant information from these sources is time-consuming, prone to errors, and inefficient.

Recent advancements in Large Language Models (LLMs) have significantly improved natural language understanding and generation capabilities. However, these models still face limitations such as hallucination and lack of domain-specific knowledge. To address these challenges, Retrieval-Augmented Generation (RAG) has emerged as an effective approach that combines information retrieval with generative models.

RAG improves the quality of LLM outputs by incorporating relevant contextual information retrieved from external knowledge sources. While traditional RAG systems rely on a single retrieval mechanism, modern enterprise applications demand more flexible and adaptive systems that can intelligently select the most appropriate source of information.

This project proposes an Adaptive Agentic RAG Framework that integrates multiple retrieval stratgies, including document-based retrieval, structured database querying, and web-based information retrieval. The system introduces a query routing mechanism that directs user queries to the most suitable knowledge source, thereby improving accuracy, contextual relevance, and scalability.

1.2 Background

1.2.1 Evolution of Information Retrieval Systems

Traditional information retrieval systems primarily relied on keyword-based approaches such as BM25, which rank documents based on term frequency and inverse document frequency. While effective for exact matching, these methods do not capture semantic relationships between queries and documents.

1.2.2 Emergence of Semantic Search

With the development of embedding models, textual data can be represented as vectors in a high-dimensional space. This enables semantic similarty search, allowing systems to retrieve contextually relevant information even when exact keywords are absent.

1.2.3 Retrieval-Augmented Generation (RAG)

RAG combines two main components:

A retriever that fetches relevant documents
A generator (LLM) that produces responses

This approach improves accuracy, contextual grounding, and reliability. However, traditional RAG systems are limited to a single retrieval pipeline and do not adapt well to varying query types.

1.2.4 Agentic AI and Workflow Orchestration

Agentic AI introduces structured workflows where tasks are broken down into smaller steps. Frameworks such as LangGraph enable conditional routing, multi-step reasoning, and modular execution, making systems more flexible and intelligent compared to static pipelines.

1.2.5 Need for Adaptive RAG Systems

Enterprise applications often require:

Handling multiple data sources
Differentiating between types of queries
Integrating structured and unstructured data

These requirements motivate the development of Adaptive Agentic RAG systems, which dynamically select the most suitable retrieval strategy.

1.3 Motivation

The motivation for this work arises from several limitations in existing document processing systems.

Key Challenges:

Large volumes of unstructured enterprise data

Inefficient manual extraction processes

Limited contextual understandng in traditional systems

Dependence on single-source retrieval in standard RAG

Difficulty in handling diverse query types

Purpose of the Project:

Automate information extraction from enterprise documents

Improve retrieval accuracy through semantic search

Enable dynamic query routing across multiple sources

Reduce dependence on manual analysis

1.4 Objectives

The primary objective of this work is to develop a Retrieval Augmented Generation based system that enhances document understanding and automated information extraction. This includes implementng semantic retrieval using vector embeddings and integrating language models for contextual response generation. In addition, the project aims to design an adaptive query routing mechanism capable of selecting appropriate knowledge sources such as vector databases, structured data stores, and web based resources to improve response relevance and reliability.

1.5 Contributions

This project contributes in the following areas:

A] Technical Contributions:

Implementation of a Retrieval-Augmented Generation pipeline

Integration of vector-based semantic retrieval

Development of an Adaptive Agentic RAG framework

Implementation of multi-source query routing

B] Research Contributions:

Study of advanced RAG strategies including:

Agentic RAG

Corrective RAG

Reflexion RAG

Self-RAG

Adaptive RAG

Mapping theoretical concepts to practical implementation

C] Practical Contributions:

Development of a system capable of:

Handling multiple query types

Integrating diverse data sources

Generating context-aware responses

1.6 Scope of the Work

The implemented scope of this project includes document ingestion, preprocessing, semantic embedding generation, vector storage, retrieval based context extraction, language model driven response generation, and adaptive query routing. Future extensions may include structured BRD generation in standard document formats, hybrid retrieval combining keyword and semantic approaches, and enhanced agent based reasoning capabilities.

2. LITERATURE REVIEW

2.1 Introduction

The advancement of Large Language Models and Retrieval-Augmented Generation has significantly influnced the development of intelligent document processing systems. This chapter reviews the evolution of retrieval techniques, RAG architectures, and agentic frameworks that contribute to modern AI-driven solutions.

The discussion progresses from traditional retrieval methods to advanced adaptive systems that integrate multiple knowledge sources and reasoning capabilities.

2.2 Traditional Information Retrieval

Early information retrieval systems relied on keyword-based techniques such as TF-IDF and BM25. These methods rank documents based on term frequency and statistical relevance.

Characteristics:

Based on exact keyword matching

Efficient for structured queries

Widely adopted in search engines

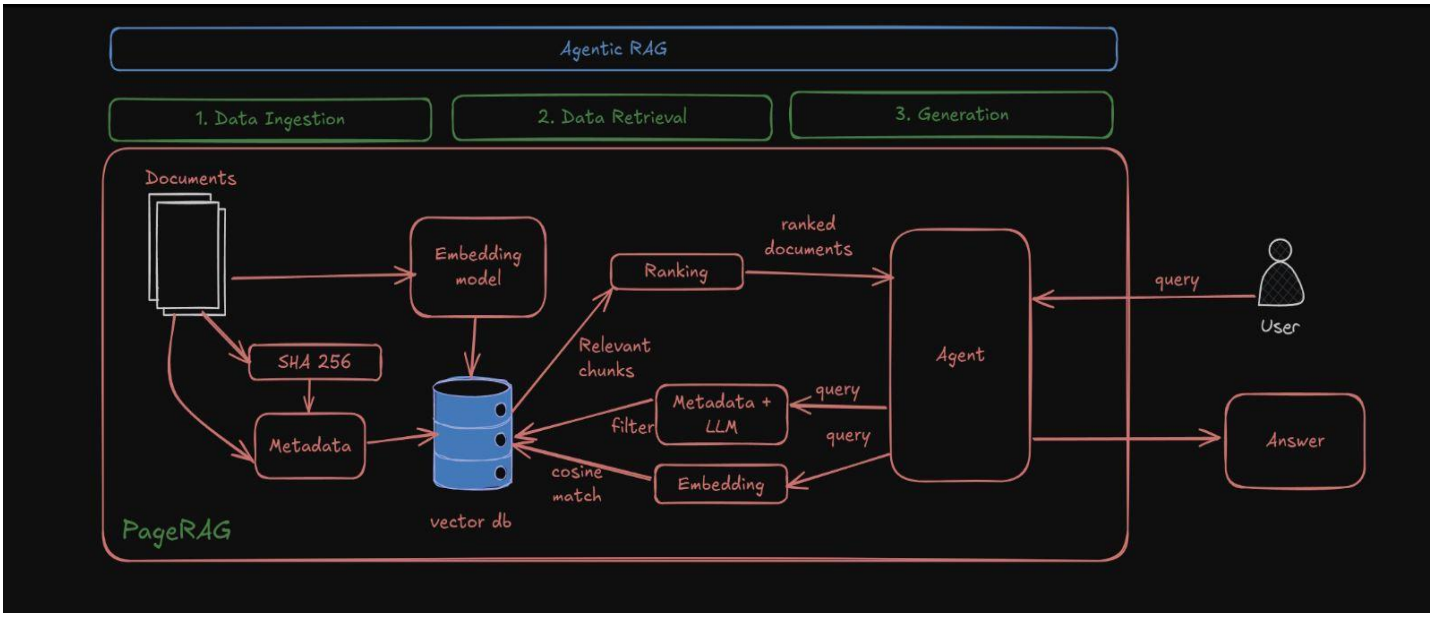
Limitations:

Lack of semantic understanding

Ineffective for paraphrased queries

Limited contextual awareness

These limitations led to the development of semantic retrieval techniques.



2.3 Semantic Retrieval and Vector Search

Semantic Retrieval and Vector Search Semantic retrieval represents a shift toward meaning based information access through the use of embedding models. By encoding text into vector representations, systems can evaluate similarity using distance measures such as cosine similarity. Vector databases provide

scalable infrastructure for storing and retrieving embeddings and have significantly improved retrieval accuracy in modern applications.

2.4 Retrieval-Augmented Generation

Retrieval-Augmented Generation combines retrieval mechanisms with generative models to improve response quality.

Retrieval Augmented Generation integrates document retrieval with natural language generation to enhance factual grounding. The process involves embedding the query, retrieving relevant content, and conditioning the language model on this information during response generation. This approach has been widely adopted to reduce hallucination and improve domain specific performance.

This approach forms the foundation for many modern AI-based document systems.

2.5 Hybrid Retrieval Approaches

Hybrid retrieval combines semantic search with keyword-based retrieval to improve performance.

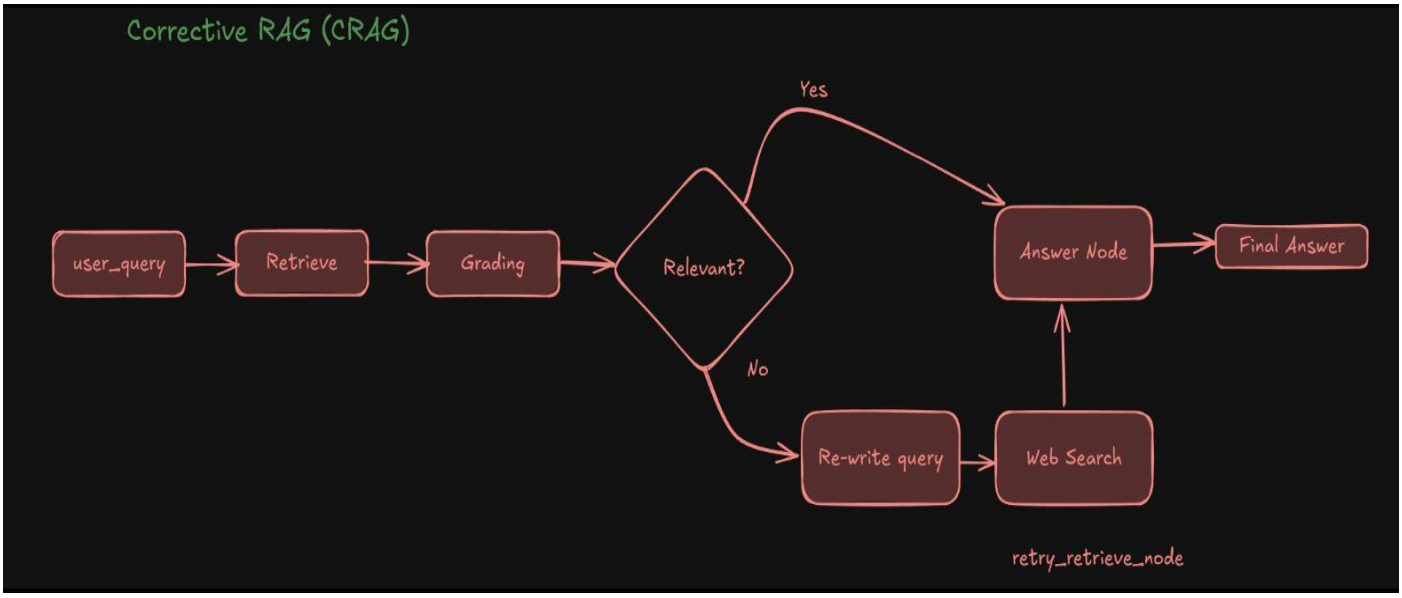
Hybrid retrieval techniques combine keyword based and semantic retrieval strategies to balance precision and contextual relevance. Such approaches are particlarly effective in technical and enterprise domains where both exact matching and semantic understanding are important.

2.6 Agentic RAG Systems

Agentic RAG systems extend traditional retrieval-augmented architectures by introducing structured workflows in which multiple components collaborate to accomplish a task. These systems typically consist of modules such as a planner, retriever, generator, and validator, each responsble for a specific function within the pipeline. Through mechanisms such as task decomposition, conditional execution, and multi-step reasoning, agentic RAG systems enable more organized and intelligent processing of queries. As a result, they offer improved flexibility and scalability compared to conventional RAG pipelines, making them suitable for complex and dynamic application scenarios.

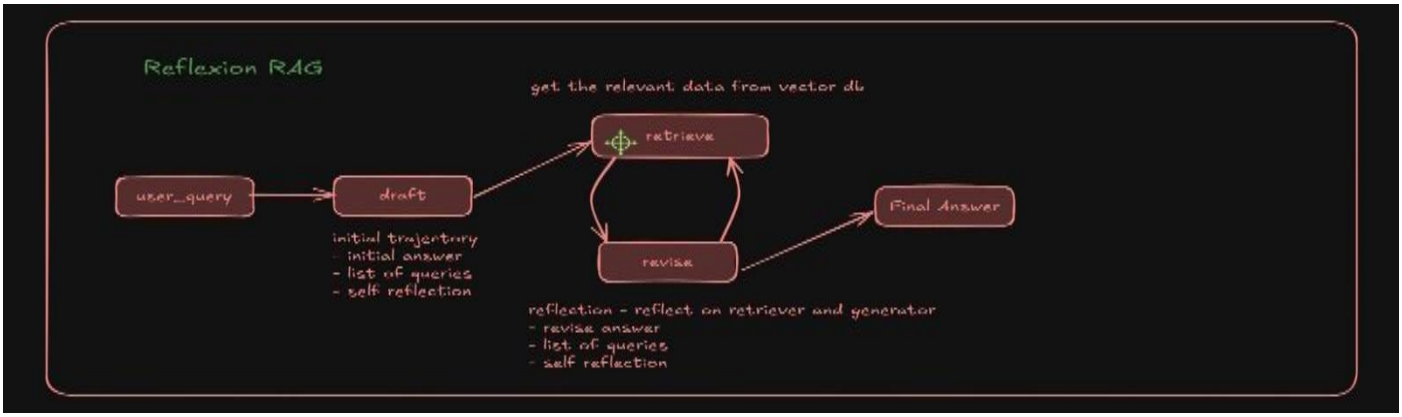
2.7 Corrective RAG

Corrective RAG enhances the reliability of retrieval-augmented systems by addressing issues related to incorrect or irrelevant document retrieval. It incorporates mechanisms to evaluate the quality of retrieved results, filter out irrelevant information, and utilize fallback strategies such as web-based retrieval when necesary. By ensuring that only relevant context is used during response generation, corrective RAG improves both the accuracy and robustness of the system.



2.8 Self-RAG and Reflexion Models

Self-RAG introduces adaptive behaviour by enabling the model to determine when external retrieval is required, rather than relying on a fixed retrieval process. In contrast, Reflexion based models incorporate feedback loops that allow the system to iteratively refine its responses. Together, these approaches support adaptive decision making, improve response quality, and reduce dependence on static pipelines. They represent a significant step toward the development of more autonomous and inteligent AI systems.

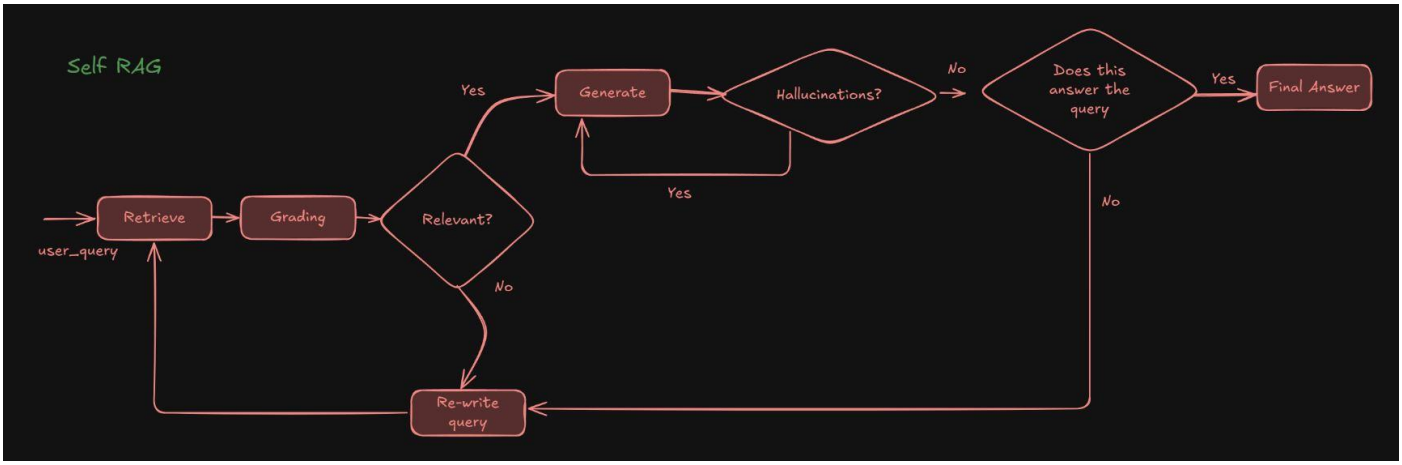


2.9 Structured Generation

Structured generation focuses on transforming unstructured input data into well-organized and meaningful formats. This approach is particularly useful in applications such as business document generation, requirement extraction, and data structuring. In enterprise environments, structured generation plays a crucial role in automating tasks like Business Requirement Document (BRD) creation, where consistency and clarity are essential.

2.10 Document Automation Using RAG

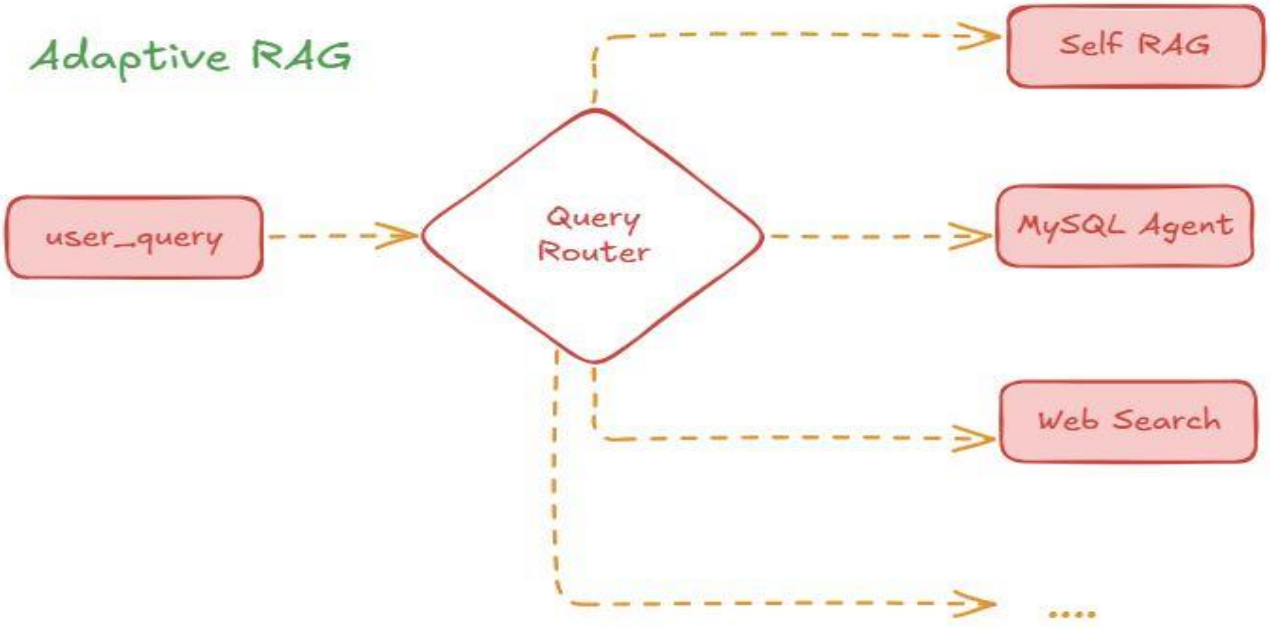
RAG-based systems have been widely adopted for document understanding and question answering tasks. These systems typically follow a pipeline consisting of document ingestion, embedding generation and storage, retrieval of relevant information, and response generation using a language model. By combining retrieval with generation, such systems effectively leverge external knowledge sources to produce accurate and context-aware outputs, demonstrating their practical applicability in real-world scenarios.



2.11 Adaptive RAG Systems

Modern Retrieval-Augmented Generation (RAG) systems are evolving toward adaptive frameworks that dynamically select retrieval strategies based on the nature of the input query. Unlike traditinal approaches that rely on a fixed pipeline, adaptive RAG systems incorporate mechanisms for query classification and multi-source routing, enabling the system to direct queries to appropriate knowledge sources such as document repositories, structured databases, or external web resources. Additionally, these systems emphasize context-aware generation, where retrieved information is effectively utilized by the language model to produce relevant and coherent responses. This adaptive paradigm forms the foundation of the

system proposed in this work, enhancing flexiblity and overall performance in handling diverse query scenarios.



3. METHODOLOGY

3.1 System Overview

The proposed system follows an Adaptive Agentic Retrieval Augmented Generation framework that is designed to process user queries in a systematic and intelligent manner. The overall architecture is organized as a modular pipeline in which each component performs a well defined role, while coordination among components is managed through an agentic execution flow. The system begins by analysing the incoming query, after which an approprate knowledge source is selected, relevant information is retrieved, and a context aware response is generated. This modular design improves transparency, simplifies debuggiing, and allows individual components to be enhanced or replaced without affecting the overall workflow.

3.2 Dataset and Input Sources

The proposed system operates on dynamically ingested data rather than relying on a fixed dataset. It supports multiple input sources, including PDF documents, Word files, text files, internal organizational databases, and web based data. This flexibility enables the system to handle diverse and continuously evolving enterprise information.

3.3 Data Preprocessing

Data preprocesing plays a crucial role in preparing raw input for effective embeding and retrieval. The process involves text extraction, data cleaning, normalization, and chunking. Chunking divides large documents into smaller, manageable segments while preserving contextual information, thereby improving retrieval accuracy and efficiency.

3.4 Model Architecture

The system architecture is composed of several interconnected components, including an embedding model for vector representation, a vector database for storage and retrieval, a language model for response generation, and an agentic workflow system that manages the overall execution and coordination of tasks.

3.5 Retrieval and Reranking

The retrieval process begins with converting the user query into an embedding, followed by a similarity search in the vector database to identify relevant information. Top K results are selected based on similarity scores. Reranking is performed implicitly through these similarity measures, ensuring that the most relevant context is used for response generation.

3.6 Adaptive Agentic RAG Framework

The proposed system introduces an adaptive mechanism that dynamicaly selects the most appropriate retrieval strategy based on the nature of the user query. This enables efficient handling of diverse queries by leveraging different knowledge sources within a unified framework.

3.7 Query Classification and Routing

The system classifies incoming queries into three categories to determine the appropriate processing route. Document-based queries are handled through vector-based retrieval combined with language model generation. Structured queries are procesed using direct database lookup, ensuring fast and accurate responses. External or out-of-domain queries are handled through web search and subsequent summarization, enabling the system to access up-to-date information.

3.8 Agentic Workflow

The agentic workflow coordinates the overall system operation by managing query classification, routing decisions, retrieval execuution, and response generation. This structured approach ensures efficient and organized processing of tasks across different components.

3.9 Integrated RAG Strategies

The system integrates multiple RAG strategies into a unified framework, combining retrieval, generation, and adaptive routing. This integration enhances the system's ability to handle diverse query types and improves overall performance.

3.10 System Workflow

The system follows a structured workflow in which a user query is first received and analyzed. The query is then classified to determine the appropriate route, after which relevant data is retrieved from the selected source. Finally, the language model generates a response based on the retrieved context.

3.11 Scope Clarification

The current implementation includes retrieval-based generation, vector-based semantic search, and adapptive query routing. However, certain aspects remain as future work, including structured document generation and the incorporation of advanced reasoning mechanisms to further enhance system capabilities.

3.12 Summary

The methodology demonstrates an adaptive system capable of handling diverse queries through intelligent routing.

4. RESULTS AND ANALYSIS

4.1 Introduction

This chapter evaluates system performance based on qualitative analysis and observed behaviour.

4.2 Evaluation Metrics

The performance of the proposed system is evaluated using a set of qualitative metrics tailored to its functionality. The primary evaluation criteria include retrieval relevance, response accuracy, effective utilizaation of contextual information, routing accuracy, and overall response coherence. In addition, secondary factors such as response time, completeness of the generated output, and user satisfaction are considered to assess the practical usability of the system.

4.3 System Performance

The system demonstrates strong performance across different query categories. For document-based queries, it achieves high accuracy with contextually relevant responses due to effective retrieval mechanisms. Structured database queries are handled efficiently, providing fast and precise results. In the case of web-based queries, the system generates informative responses, although the quality may depend on the reliability of external data sources.

4.4 Query Routing Analysis

The query routing mechanism is effective in classifying and directing most queries to the appropriate processing module. However, certain ambiguous queries may occasionally lead to suboptimal routing decisions, which can impact the overall response quality.

4.5 Qualitative Analysis

The system demonstrates consistent performance in generating context-aware responses across a variety of query types. The integration of retrieval and generation enables the production of meaningful and coherent outputs.

4.6 Retrieval Analysis

The effectiveness of the retrieval process depends on several factors, including chunk size, the quality of embeddings, and the indexing strategy used in the vector database. Proper tuning of these parameters is essential for achieving optimal performance.

4.7 Efficiency

In terms of efficiency, database queries exhibit the lowest response time due to direct lookup mechanisms. In contrast, web based queries tend to have higher latency because of external data fetching and processing.

4.8 Comparative Analysis

Compared to traditional RAG systems, the proposed framework offers improved adaptablity and flexibility by incorporating multiple retrieval strategies and dynamic query routing.

4.9 Summary

Overall, the system effectively handles diverse query types and enhances response quality through adaptive routing and multi-source integration.

5. DISCUSSION

5.1 Overview

This chapter presents a discussion of the system's behaviour based on observations obtained during implementation and evaluation. It highlights key insights, strengths, and limitations of the proposed framework.

5.2 Key Observations

The performance of the system is strongly influenced by the quality of retrieval, as accurate and relevant context directly impacts the generated responses. The adaptive routing mechanism enhances system flexibility by enabling appropriate handling of different query types. Additionaly, the integration of multiple knowledge sources contributes to improved robustness and overall system reliability.

5.3 Advantages

The proposed system supports a wide range of data sources, including both structured and unstructured formats. It is capable of generating context-aware responses by effectively utilizing retrieved information. Furthrmore, the modular and scalable architecture allows the system to be extended for larger and more complex applications.

5.4 Limitations

Despite its advantages, the system has certain limitations. It does not currently support structured output generation, such as automated document creation. The implementation of advanced reasoning mechanisms remains limited, and the system's performance is dependent on the quality and relevance of the input data.

5.5 Summary

The system demonstrates strong potential in handling diverse queries and generating meaningful responses. However, further enhancements are required to improve its capablities and extend its applicability.

6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presents an Adaptive Agentic RAG framework that enhances information retrieval and response generation through the integration of multiple knowledge sources and dynamic query routing. The system effectively demonstrates how combining retrieval, generation, and adaptive decision-making can improve the performance and flexibility of AI-based document processing systems.

6.2 Future Work

Future work can focus on extending the system to support structured document generation, such as automated BRD creation. Additional improvements may include the incorporation of hybrid retrieval techniques, the development of more advanced agentic workflows, and enhancements in query classification mechanisms to further improve accuracy and efficiency.

6.3 Final Remark

The proposed framework provides a solid foundation for future research and practical deployment of intelligent document automation systems. Its adaptive and modular design positions it as a promising solution for evolving enterprise information processing requirements.